Expanding SEPSES Cybersecurity Knowledge Graphs (CSKG)

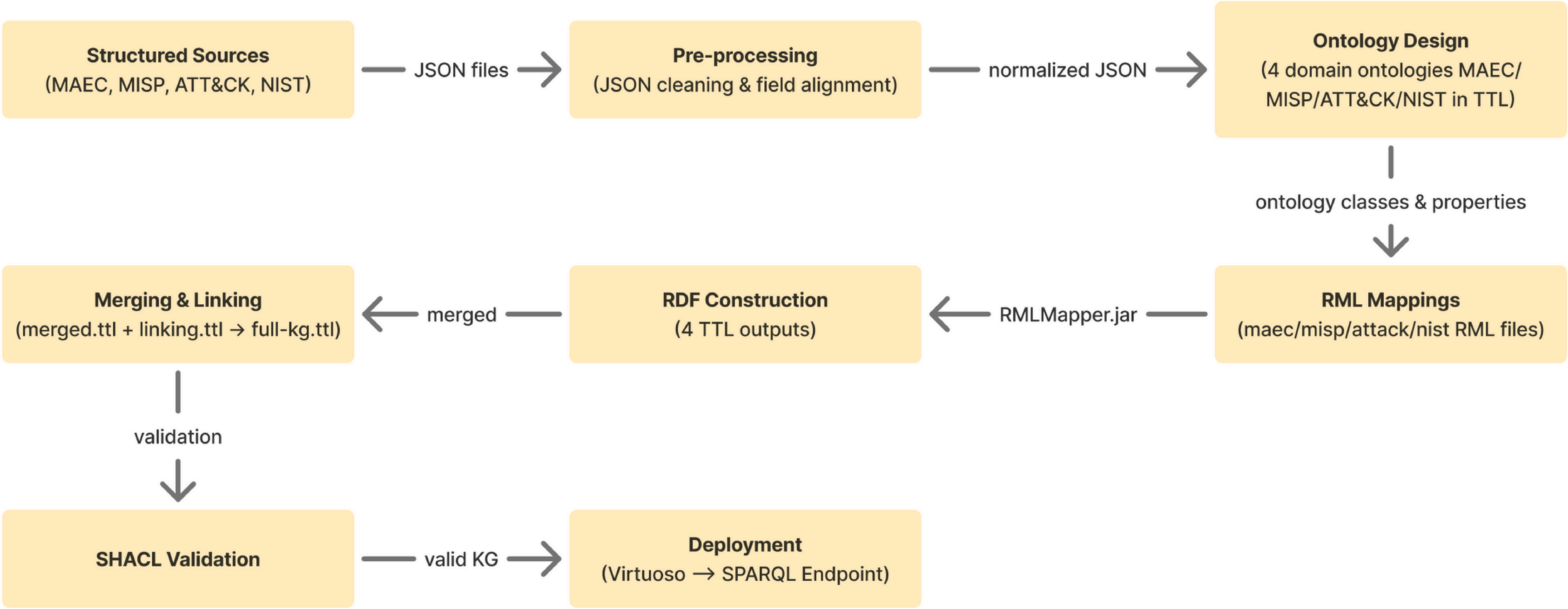
Kelompok 1

- Alifa Batrisyia Nariswari (24/535256/PA/22707)
- Ananda Auliya Rahma (24/533691/PA/22608)
- Kesya Izumi (24/533729/PA/22612)
- Aulia Fathus Tsani (24/534388/PA/22661)
- Herlina Iin Nur Soleha (24/541333/PA/22962)

Description

- Menambahkan **minimal 3 sumber data** keamanan siber terstruktur baru dan menentukan atau **membuat ontologi** yang sesuai.
- **Mengembangkan pipeline (RML Mapping)** untuk membangun CSKG dari sumber-sumber tersebut.
- **Menghubungkan hasil Knowledge Graph** baru dengan entitas yang sudah ada di SEPSES CSKG.
- Menyajikan **ringkasan**, **evaluasi**, dan minimal **3 use cases** dari Knowledge Graph yang dibuat.
- Menyimpan hasil dalam format RDF/Turtle, mengunggahnya ke SPARQL endpoint (Virtuoso/QLever), serta menyediakan semua hasil dan kode di GitHub.

Workflow Overview



I. Data Sources

4 data publik dalam format JSON/OSCAL/STIX:

1. **NIST SP 800-53 (OSCAL JSON)**: https://github.com/usnistgov/oscal-content/blob/main/nist.gov/SP800-53/rev5/json/NIST_SP-800-53_rev5_catalog-min.json

2. **MISP Feed Metadata**: https://github.com/MISP/MISP/blob/2.4/app/files/feed-metadata/defaults.json

3. **MAEC Malware Action Schema**: https://github.com/MAECProject/schemas/blob/master/malware-action.json

4. **MITRE ATT&CK (STIX JSON)**: https://github.com/mitre-attack/attack-stix-data/blob/master/mobile-attack/mobile-attack-1.0.json

II. Ontology Design

1. **ATT&CK Ontology**
   Technique, AdversaryGroup, usesTechnique

Listing 1.1: ATT&CK Ontology in TTL Format

```
@prefix attack: <http://w3id.org/sepses/vocab/ref/attack#> .

attack:AdversaryGroup rdf:type owl:Class ;
    rdfs:label "AdversaryGroup" .

attack:Technique rdf:type owl:Class ;
    rdfs:label "Technique" .

attack:usesTechnique rdf:type owl:ObjectProperty ;
    rdfs:domain attack:AdversaryGroup ;
    rdfs:range attack:Technique .

attack:techniqueID rdf:type owl:DatatypeProperty ;
    rdfs:range xsd:string .
```

2. **MAEC Ontology**
   MalwareAction, APICall, hasAPICall

Listing 1.2: MAEC Ontology in TTL Format

```
@prefix maec: <http://w3id.org/sepses/vocab/ref/maec#> .

maec:MalwareAction rdf:type owl:Class ;
    rdfs:label "Malware␣Action" ;
    rdfs:comment "A␣single␣atomic␣action␣performed␣by␣malware." .

maec:APICall rdf:type owl:Class ;
    rdfs:label "API␣Call" .

maec:hasAPICall rdf:type owl:ObjectProperty ;
    rdfs:domain maec:MalwareAction ;
    rdfs:range maec:APICall .

maec:actionName rdf:type owl:DatatypeProperty ;
    rdfs:range xsd:string .
```

II. Ontology Design

3. MISP Ontology

Feed, Event, Attribute, hasIndicator

Listing 1.3: MISP Ontology in TTL Format

```
@prefix misp: <http://w3id.org/sepses/vocab/ref/misp#> .
@prefix mispf: <http://w3id.org/sepses/vocab/ref/misp/feed/> .
@prefix schema: <http://schema.org/> .
@prefix dct: <http://purl.org/dc/terms/> .
@prefix skos: <http://www.w3.org/2004/02/skos/core#> .

mispf:Feed rdf:type owl:Class ;
   rdfs:label "MISP␣Feed" .

misp:Event rdf:type owl:Class ;
   rdfs:label "MISP␣Event" .

misp:hasIndicator rdf:type owl:ObjectProperty ;
   rdfs:domain misp:Event ;
   rdfs:range misp:Attribute .

mispf:feedName rdf:type owl:DatatypeProperty ;
   rdfs:subPropertyOf schema:name .
```

4. NIST Ontology

Control, Profile, Metadata

Listing 1.4: NIST Ontology in TTL Format

```
@prefix nist: <http://w3id.org/sepses/vocab/ref/nist#> .

nist:Control rdf:type owl:Class ;
   rdfs:label "Control" .

nist:Profile rdf:type owl:Class ;
   rdfs:label "Profile" .

nist:Metadata rdf:type owl:Class ;
   rdfs:label "Metadata" .

nist:hasMetadata rdf:type owl:ObjectProperty ;
   rdfs:domain nist:Profile ;
   rdfs:range nist:Metadata .

nist:controlID rdf:type owl:DatatypeProperty ;
   rdfs:range xsd:string .
```

III. KG Construction Pipeline (RML Mapping)

Menggunakan **RMLMapper.jar**

```
java -jar rmlmapper.jar -m <mapping-file>.ttl -o <output-file>.ttl
```

Setiap mapping memiliki:

- rr:LogicalSource → sumber JSON
- rr:SubjectMap → URI resources
- rr:PredicateObjectMap → mapping field ke ontology

Hasil mapping (KG):

- attack_output.ttl
- maec_output.ttl
- misp_output.ttl
- nist_output.ttl

IV. Linking Strategy

1. **Vocabulary Alignment**

   SKOS, Schema.org, DCTerms
2. **Semantic Linking**
   - MISP Event → Attribute
   - Attribute → MAEC / ATT&CK
   - MAEC → NIST
3. **Unified Graph**

   merged.ttl + linking.ttl → full-kg.ttl
   - merged.ttl : gabungan (union) dari keempat output RDF
   - linking.ttl : penyelarasan kosakata dan penghubungan antar model
   - full-kg.ttl : CSKG terintegrasi final

   Knowledge graph final (full-kg.ttl) berisi:
   - Seluruh kelas dan properti ontologi
   - Semua instance yang dihasilkan dari sumber input JSON

Use Cases Applications

1. **Use Case 1: Threat Traceability**

   End-to-end chain:

   MISP Feed → Event → Attribute → MAEC Malware Action → MITRE ATT&CK Technique → NIST Privacy Control

```
PREFIX misp: <http://w3id.org/sepses/vocab/ref/misp#>
PREFIX mispf: <http://w3id.org/sepses/vocab/ref/misp/feed/>
PREFIX maec: <http://w3id.org/sepses/vocab/ref/maec#>
PREFIX attack: <http://w3id.org/sepses/vocab/ref/attack#>
PREFIX nistp: <http://w3id.org/sepses/vocab/ref/nist-privacy#>
SELECT DISTINCT ?feed ?event ?attribute ?maec ?technique ?control
WHERE {
  ?feed a misp:MISPFeed .
  ?feed misp:producesEvent ?event .
  ?event misp:hasIndicator ?attribute .
  OPTIONAL { ?attribute misp:linkedMAEC ?maec . }
  OPTIONAL { ?attribute misp:linkedAttack ?technique . }
  OPTIONAL { ?maec maec:impactsPrivacyControl ?control . }
}
LIMIT 200
```

Use Cases Applications

2. **Use Case 2: Gap Analysis**

   Mengidentifikasi kekurangan dalam KG:

   - **Events without attributes**

```
PREFIX misp: <http://w3id.org/sepses/vocab/ref/misp#>
SELECT ?event
WHERE {
  ?event a misp:Event .
  FILTER NOT EXISTS { ?event misp:hasIndicator ?attr }
}
```

   - **MAEC actions without mapping to ATT&CK**

```
PREFIX maec: <http://w3id.org/sepses/vocab/ref/maec#>
SELECT ?maec
WHERE {
  ?maec a maec:MalwareAction .
  FILTER NOT EXISTS { ?maec maec:mapsToAttack ?t }
}
```

   - **Attributes without MAEC/ATT&CK links**

```
PREFIX misp: <http://w3id.org/sepses/vocab/ref/misp#>
SELECT ?attr
WHERE {
  ?attr a misp:Attribute .
  FILTER NOT EXISTS { ?attr misp:linkedMAEC ?m }
  FILTER NOT EXISTS { ?attr misp:linkedAttack ?t }
}
```

Use Cases Applications

3. Use Case 3: Quantitative Analysis

Three metrics are evaluated:

- **Event Count per Feed**

```
PREFIX misp: <http://w3id.org/sepses/vocab/ref/misp#>
SELECT ?feed (COUNT(?event) AS ?numEvents)
WHERE {
  ?feed a misp:MISPFeed .
  ?feed misp:producesEvent ?event .
}
GROUP BY ?feed
```

- **ATT&CK Techniques per NIST Privacy Control**

```
PREFIX attack: <http://w3id.org/sepses/vocab/ref/attack#>
PREFIX nistp: <http://w3id.org/ns/sepses/vocab/ref/nist-privacy#>
SELECT ?control (COUNT(?technique) AS ?numTechniques)
WHERE {
  ?technique attack:hasPrivacyImpact ?control .
}
GROUP BY ?control
ORDER BY DESC(?numTechniques)
```

- **MAEC actions per ATT&CK techniquel**

```
PREFIX misp: <http://w3id.org/sepses/vocab/ref/misp#>
SELECT ?feed (COUNT(?event) AS ?numEvents)
WHERE {
  ?feed a misp:MISPFeed .
  ?feed misp:producesEvent ?event .
}
GROUP BY ?feed
```

Statistic Summary and Evaluation

1. **Triple Count**

Table 1: Triple Counts per Source and Final Integrated Graph

Source Ontology	Triples	Notes
MITRE ATT&CK	7,575	Largest contributor; rich behavior relationships
MAEC	99	Small ontology; high granularity, low volume
MISP Feed	1,656	Metadata-rich, moderate size
NIST Controls	361	High data property density
Full Integrated KG	**14,926**	Includes linking triples across ontologies

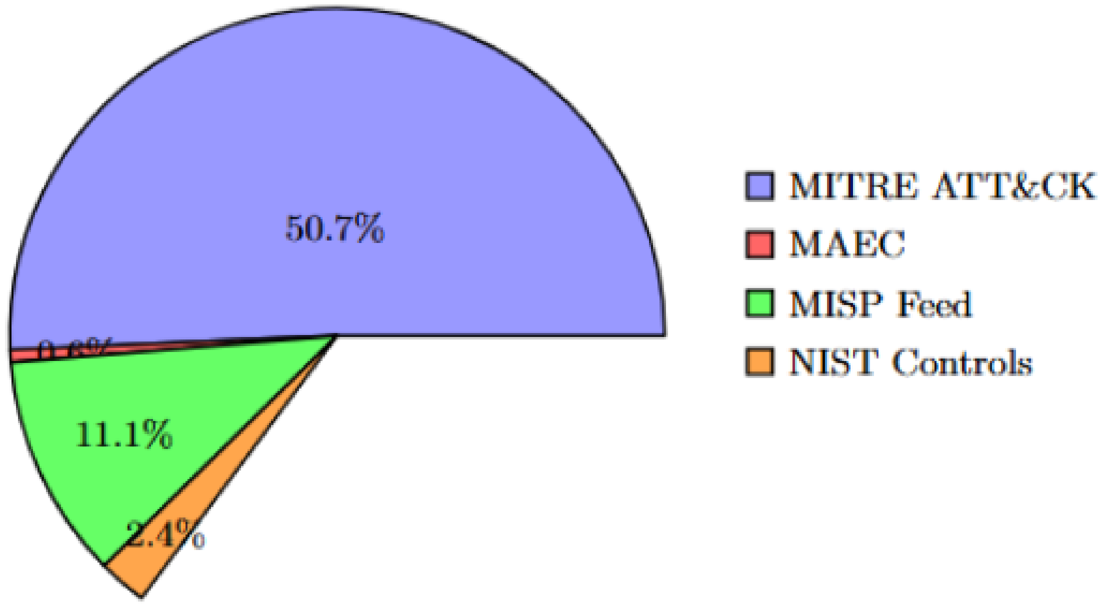


Fig. 2: Percentage Contribution of Triples Across Ontology Sources

Statistic Summary and Evaluation

2. Ontology Metric Summary

Table 2: Structural Metrics for Each Ontology

Ontology	Classes	Object Properties	Data Properties	Annotation Properties
MITRE ATT&CK	14	15	10	17
MAEC	3	2	13	23
MISP Feed	4	2	11	72
NIST Controls	7	5	17	40

3. SHACL Validation

Table 3: SHACL Validation Results

Validation Target	SHACL Result	Violations
ATT&CK Output	Conforms	0
MAEC Output	Conforms	0
MISP Output	Conforms	0
NIST Output	Conforms	0
Full KG (Merged)	**Conforms**	**0**

Statistic Summary and Evaluation

4. **Evaluation**

   Although SHACL validation showed no violations and all required properties are populated, several limitations remain:

   - Coverage imbalance – ATT&CK dominates, making the KG behavior centric.
   - Limited MAEC footprint – Its small ontology contributes few triples.
   - Source dependency – MISP feeds and NIST control data heavily rely on source quality.
   - No temporal reasoning – Time-based relationships are not modeled.
   - Limited inference – Only structural validation was performed; reasoning based validation was not.

Attachments

1. **Semua file ontology disimpan di GitHub:**
   https://github.com/Aria1516/Kelompok-1_CSKG

2. **Final Report:**
   https://www.overleaf.com/project/691d329c8cba3d59aa339ce3

Terima Kasih

LOCALLY ROOTED,
GLOBALLY RESPECTED

ugm.ac.id